AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Final Examination
Spring Semester 2025/2026

Date: 4/June/2026

AOU EGYPT

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

1. What is the correct ordering of algorithmic complexities from best (most efficient) to worst (least efficient)?
   - A) $O(n^2)$, $O(n^2 \log n)$, $O(n \log n)$, $O(n)$
   - B) $O(n)$, $O(n \log n)$, $O(n^2)$, $O(n^2 \log n)$
   - C) $O(n \log n)$, $O(n)$, $O(n^2 \log n)$, $O(n^2)$
   - D) $O(n)$, $O(n^2)$, $O(n \log n)$, $O(n^2 \log n)$
2. Which recurrence relation represents the complexity of a binary search?
   - A) $T(n)=2T(n/2)+k$
   - B) $T(n)=T(n-1)+k$
   - C) $T(n)=T(n/2)+k$
   - D) $T(n)=T(n/4)+k$
3. In graph theory, if m represents the number of edges, what is the sum of the degrees of all vertices?
   - A) m
   - B) m^2
   - C) $2m$
   - D) $m/2$
4. Which data structure is the most appropriate for representing relationships between friends in a social network?
   - A) Tree
   - B) Graph
   - C) Max-Heap
   - D) Sorted Array
5. What is the average number of searches required to find an item in a sorted array of 32 items using binary search?
   - A) 4
   - B) 5
   - C) 16
   - D) 32
6. Which searching algorithm is faster than a naive linear search?
   - A) Breadth-First Search
   - B) Brute-force MST
   - C) KMP (Knuth-Morris-Pratt)
   - D) Depth-First Search
7. How is the Depth-First Search (DFS) algorithm characterized?
   - A) It explores as deep as possible along each branch before backtracking.
   - B) It processes all of a vertex's descendants before moving to an adjacent vertex.
   - C) It belongs to the divide and conquer paradigm.
   - D) It is a linear data structure.
8. What occurs when searching for a value that is NOT available in a list?
   - A) Best case performance
   - B) Average case performance
   - C) Worst case scenario
   - D) Logarithmic performance
9. How are NP problems defined in terms of complexity?
   - A) Problems that are easy to solve but hard to verify.
   - B) Problems with yes/no answers that are easy to verify but hard to solve.
   - C) Problems that always follow the $O(n \log n)$ complexity.
   - D) Problems that can only be solved using the brute-force paradigm.
10. What is the complexity of the best-known algorithm to find a pair of numbers 'a' and 'b' in a sorted array of n numbers such that the absolute difference $|a-b|$ equals a positive integer k?
    - A) $O(\log n)$
    - B) $O(n)$
    - C) $O(n \log n)$
    - D) $O(n^2)$

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PART 2: ALL QUESTIONS ARE REQUIRED [40 Marks]

Question 2: (15 Marks)

a) Identify the type of logic described in the following phrases: **(4 marks: 2 marks each)**

1) A formal language that uses variables and quantifiers (such as "for all" or "there exists") to express logical schemes.
2) A branch of logic that focuses on combining or modifying entire sentences or statements to create complex logical properties.

b) Define the following classes of problems: **(4 marks: 2 marks each)**

1) Problems that a computer can solve efficiently because they have straightforward yes-or-no answers.
2) Problems where a computer can quickly verify if a given solution is correct, but finding the solution itself is extremely time-consuming.

c) Identify the algorithm paradigm described and provide one example of a famous algorithm belonging to that paradigm: **(4 marks: 2 marks each)**

1) This approach divides a problem into smaller, independent parts, solves each individually, and then combines the results.
2) This approach makes the locally optimal choice at each step, hoping to find the overall optimal solution.

d) Determine the complexity of an algorithm described by the following recurrence relation. $T(n)=9T(n/3)+n^2$ **(3 marks)**

Question 3: (15 Marks)

a) Given the following List Arr = {25, 45, 19, 78, 12, 34, 22, 51, 27}

1) Build a binary search tree for the list Arr. **(2 marks)**
2) Assuming that the binary search tree is balanced, what is the complexity of finding a specific element in the tree? **(2 marks)**
3) Redraw the binary search tree after removing the item 25. **(2 marks)**
4) Insert the original list Arr in a hash-table data structure using hash function $h(i)=(2i+1)\text{mod}11$, using chaining technique to handle any collision. **(4 marks)**

b) The Knuth-Morris-Pratt's (KMP) algorithm searches for a pattern inside a string. Assume that the pattern is "*abcdabcy*", and the string is "*abcdabcdabcy*".

1) Compute the failure function for the pattern. **(2 marks)**
2) Draw a figure to illustrate the comparisons that will take place till we find the pattern inside the string. **(3 marks)**

Question 4: (10 Marks)

Compute the complexity of the following tasks: **(10 marks: 1 mark each)**

1) Traverse a graph using Depth First Search algorithm, where V is the number of vertices and E is the number of edges in the graph
2) The splitting part in the Merge-Sort algorithm
3) Build the Heap
4) T(n) = 8 T(n/2)+ n
5) Finding the largest number of a sorted doubly linked list.
6) Implementing Selection-Sort algorithm on a sorted list.
7) Using recursive algorithm to compute the nth term in a Fibonacci series
8) Using dynamic algorithm to compute the nth term in a Fibonacci series
9) Searching for an element using hash tables with chaining
10) Constructing an adjacency matrix for a graph of order *n* and size *m*

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PART 3: ALL QUESTIONS ARE REQUIRED [50 Marks]

Question 5: (40 marks)

Consider the weighted graph illustrated in Fig.1

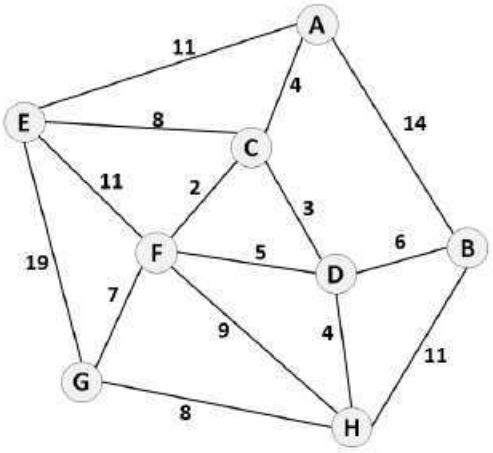


Figure 1 weighted graph

a) Does this graph contain Euler circuit or/and Euler path? Why? **(4 marks)**

b) What is the sequence of visited node in BFS, and DFS if both algorithms start from node A? **(4 marks)**

c) Draw the minimum spanning tree of this graph. **(6 marks)**. What is the name and complexity of such algorithm? **(4 mark)**

d) Use the Dijkstra's algorithm to determine the shortest paths from node (A) to all other cities **(8 marks)**. Determine the shortest path and cost from node A to node H **(2 marks)**. [Hint: use the Dijkstra's table to implement the algorithm step by step to show which node will be added in sequence]

e) Implement the proper algorithm to find the shortest paths between all pairs of nodes [Compute D_0, D_1, D_2]. Mention the name of the algorithm that could be used and its complexity **(12 marks)**

Question 6: (10 marks)

You are given a sorted list of distinct positive numbers representing recorded data values. To detect simple and meaningful patterns, analysts focus on sequences that increase by a fixed and known amount, called the common difference.
A sequence follows an arithmetic progression if the difference between consecutive values is constant. Given a specific common difference d, your task is to efficiently examine the list and print every group of three numbers that follow this pattern:

(x, x+d, x+2d)

Since the list may be large, the solution should avoid unnecessary comparisons and run faster than a quadratic-time approach.

	Input	**Output**
Example 1	A[] = { 6, 9, 10, 12, 13, 16 }, d = 3	6 9 12 10 13 16
Example 2	A[] = { 3, 5, 7, 8, 10, 11, 17 }, d = 2	3 5 7

a) Design an efficient algorithm to solve this problem. (**7 marks**) [The efficiency of your algorithm is the main driver of the mark],

b) Analyze the complexity of your solution. (**3 marks**) [full explanation of your answer should be provided]

End of questions

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